

Why Convex Optimization?

since other optimizations are hard to solve. convexity provides basic assumptions.

Convex Set.

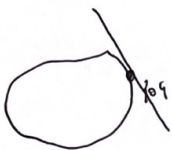
- Separating hyperplane theorem.

C, D nonempty, disjoint, convex. $a^T x \leq b, \forall x \in C, a^T x \geq b, \forall x \in D$, for $\exists a \neq 0, b$.

pf. i) $C-D$ is convex. set not containing 0.

trivial.

ii) Case I $d(C-D) \neq \{0\}$.



There exists $a \in \mathbb{R}^d$ s.t. $a^T x \leq 0, \forall x \in (C-D)$

Lemma. S be closed, convex set. If $S \cap (-S) = \emptyset$ or $\{0\}$, then there exists

$a \neq 0$ s.t. $a^T x \leq 0$ for all $x \in S - \{0\}$.

pf Spherical Projection of S to S^{d-1} .



then $A \cap (-A) = \emptyset$.

$A \cup (-A) \neq S^{d-1}$ since then, contradicts to connectivity.

$\exists p \in S^{d-1}, p \notin A \cup (-A)$.

Now consider $S^{d-1} \cap \{x \mid p^T x = 0\} = S^{d-2}$.

projecting A and $-A$ to S^{d-2} , by the same logic. there is

$q \notin (\text{projection of } A) \cup (\text{projection of } -A)$.

great circle containing p and q . is the one we are finding $\square$.

iii) Case II. $d(C-D) = \{0\}$.

$\{0\}$ singleton and $d(C-D)$ convex, closed. easy version of separating hyperplane thm can be proven.

- Supporting hyperplane thm.

$\{x \mid a^T x = a^T x_0\}$, where $x_0 \in \text{bd}(C)$, $a^T x \leq a^T x_0$ for all $x \in C$.

Convex Functions.

- affine $ax+b$
- exp. e^x
- power x^α , $\alpha \geq 1$ or $\alpha \leq 0$. / $-x^\alpha$, $0 \leq \alpha \leq 1$, $x > 0$.
- negative entropy $x \log x$, $x > 0$.
- negative logarithm $-\log x$, $x > 0$.
- norms: $x \in \mathbb{R}^d$, $\|x\|$.

Techniques: restricting to line, $f(x) = \log \det(X)$, p.s.d X , is concave.

pf $\log \det(X+tV) = \log \det X + \sum_{i=1}^n \log(1+t\lambda_i)$, λ_i eigenvalues of $X^{-1/2} V X^{-1/2}$.

first order or second order conditions, log-sum-exp. is convex.

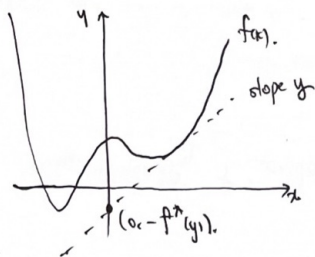
pf $f(x) = \log \sum_{k=1}^n e^{x_k}$, $\nabla^2 f(x) = \frac{1}{1^T z} \text{diag}(z) - \frac{1}{(1^T z)^2} z z^T$ ($z_k = \exp x_k$).

$$v^T \nabla^2 f(x) v = \frac{(\sum_k z_k v_k^2)(\sum_k z_k) - (\sum_k z_k v_k)^2}{(\sum_k z_k)^2} \geq 0.$$

Operations: ① Pointwise sup: $f(x,y)$ convex in x for each $y \in \mathcal{Y}$, then $\sup_{y \in \mathcal{Y}} f(x,y)$ is convex.

② Minimization: $f(x,y)$ convex, C convex set then $\inf_{y \in C} f(x,y)$ convex.

Conjugate $f^*(y) = \sup_{x \in \text{dom } f} (y^T x - f(x))$



Duality..

minimize $f_0(x)$
subject to $f_i(x) \leq 0$, $h_i(x) = 0$.

$$\Rightarrow L(x, \lambda, \mu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \mu_i h_i(x).$$

$$g(\lambda, \mu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \mu)$$

dual prob: maximize $g(\lambda, \mu)$ subject to $\lambda \geq 0$.

Weak duality: $p^* \geq d^*$

E.g. Least norm of linear equations.

minimize $x^T x$

$$\rightarrow L(x, \mu) = x^T x + \mu^T (Ax - b)$$

subject to $Ax = b$.

$$g(\mu) = -\frac{1}{4} \mu^T A A^T \mu - b^T \mu.$$

minimize $f_0(x)$

subject to $f_i(x) \leq 0 \quad i=1, \dots, m$

$h_i(x) = 0 \quad i=1, \dots, p$

$$L(x, \lambda, \mu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \mu_i h_i(x)$$

$$g(\lambda, \mu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \mu) \quad ; \text{concave.}$$

Lagrange dual problem: minimize $g(\lambda, \mu)$
subject to $\lambda \geq 0$. d^* .

$d^* = p^*$? $\leadsto$ "Constraint qualifications".

Slater's constraint Qualification.

If $\exists x \in \text{relint } \mathcal{D}$, st $f_i(x) < 0 \quad i=1, \dots, m$ and $Ax=b$. then

minimize $f_0(x)$

subject to $f_i(x) \leq 0 \quad i=1, \dots, m$

$Ax=b$

is satisfying constraint qualification.

Moreover, if some f_i 's are affine, then λ need to satisfy $f_i(x) \leq 0$ for those functions.

E.g. Least Norm of linear equations.

minimize $\|x\|_2$

subject to $Ax=b$.

strong duality holds.

$$L(x, \mu) = x^T x + \mu^T (Ax - b). \quad g(\mu) = -\frac{1}{4} \mu^T A A^T \mu + b^T \mu.$$

E.g. Minimum volume covering ellipsoid.

minimize $\log \det X^{-1}$

subject to $a_i^T X a_i \leq 1 \quad i=1, \dots, m$. ; affine constraint.

$\therefore$ strong duality holds if feasible.

$$L(x, \lambda) = \log \det X^{-1} + \sum_{i=1}^m \lambda_i [\text{tr}(a_i a_i^T X) - 1]$$

$$g(\lambda) = \log \det \left(\sum_{i=1}^m \lambda_i a_i a_i^T \right) - 1^T \lambda + n. \quad \text{if } \sum_{i=1}^m \lambda_i a_i a_i^T \succ 0.$$

Intuition: $\mathcal{S} = \{ (f_1(x), \dots, f_m(x), h_1(x), \dots, h_p(x), f_0(x)) : x \in \mathcal{D} \}$

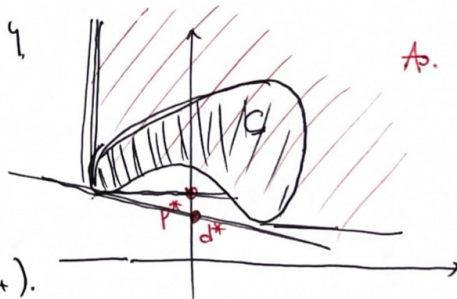
$$p^* = \inf \{ t \mid (u, v, t) \in \mathcal{S}, u \leq 0, v = 0 \}.$$

$$g(\lambda, \mu) = \inf \{ (\lambda, \mu, 1)^T (u, v, t) \mid (u, v, t) \in \mathcal{S} \}.$$

$\rightarrow$ Think of epigraph version. $\mathcal{A} = \mathcal{S} + (\mathbb{R}_+^m \times \{0\} \times \mathbb{R}_+)$.

"worse f_0 (so is larger) or worse f_i (larger)".

then $p^* = \inf \{ t \mid (0, 0, t) \in \mathcal{A} \}$. and $g(\lambda, \mu) = \inf \{ (\lambda, \mu, 1)^T (u, v, t) \mid (u, v, t) \in \mathcal{A} \}$ for $\lambda \geq 0$.



Proof of Slater's constraint qualification.

For simplicity assume $\text{relint } \mathcal{D} = \text{int } \mathcal{D}$. and $Ax=b$ constraint satisfies $\text{rank } A=p$.

Since $f_0, f_1, \dots, f_m$ are convex. $\mathcal{A}$ is convex.

; $f_2(x_1) \leq u_{21}$, $f_0(x_1) \leq t_1$. and $f_2(x_2) \leq u_{22}$, $f_0(x_2) \leq t_2$ then.

for $(\lambda u_1 + (1-\lambda)u_2, \lambda t_1 + (1-\lambda)t_2)$.

$f_i(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f_i(x_1) + (1-\lambda)f_i(x_2) \leq \lambda u_{i1} + (1-\lambda)u_{i2}$. and so does for f_0 .

and $\mathcal{B} = \{(0,0,s) \mid s < p^*\}$. $\mathcal{A} \cap \mathcal{B} = \emptyset$.

separating hyperplane thm, $\exists (\tilde{\lambda}, \tilde{\nu}, \mu) \neq 0$. $\left\{ \begin{array}{l} \tilde{\lambda}^T u + \tilde{\nu}^T v + \mu t \geq \alpha \text{ for } (u,v,t) \in \mathcal{A}. \\ \tilde{\lambda}^T u + \tilde{\nu}^T v + \mu t \leq \alpha \text{ for } (u,v,t) \in \mathcal{B}. \end{array} \right.$

By def of $\mathcal{A}$, $\tilde{\lambda} \geq 0$, $\mu \geq 0$. and by def of $\mathcal{B}$, $\mu t \leq \alpha$ for $t < p^* \Rightarrow \mu p^* \leq \alpha$.

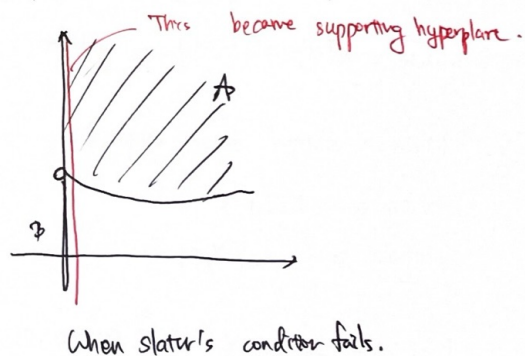
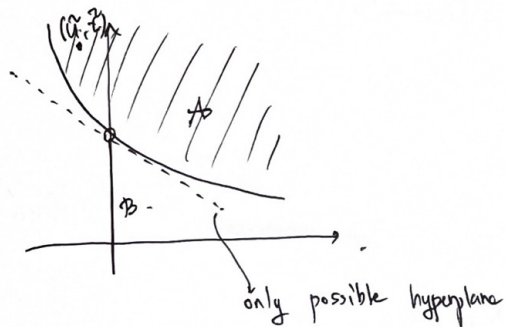
Thus, $\sum_{i=1}^m \tilde{\lambda}_i f_i(x) + \tilde{\nu}^T (Ax-b) + \mu f_0(x) \geq \alpha \geq \mu p^*$

i) If $\mu \neq 0$, $L(x, \tilde{\lambda}/\mu, \tilde{\nu}/\mu) \geq p^* \Rightarrow d^* = p^*$

ii) If $\mu = 0$, $\sum_{i=1}^m \tilde{\lambda}_i f_i(x) + \tilde{\nu}^T (Ax-b) \geq 0$.

for $\tilde{\lambda}$ satisfying Slater's condition, $\sum_{i=1}^m \tilde{\lambda}_i f_i(x) \geq 0 \Rightarrow \tilde{\lambda}_i = 0$.

thus $\tilde{\nu} \neq 0$. and so $\tilde{\nu}^T (Ax-b) \geq 0$. : contradiction. ($\because A$ is of full rank $x \in \text{int } \mathcal{D}$).



Saddle point interpretation.

Just assume inequality constraints.

$$\sup_{\lambda \geq 0} L(x, \lambda) = \begin{cases} f(x) & f_i(x) \leq 0 \quad i=1, \dots, m \\ \infty & \text{otherwise} \end{cases} \quad \therefore p^* = \inf_x \sup_{\lambda \geq 0} L(x, \lambda).$$

whereas $d^* = \sup_{\lambda \geq 0} \inf_x L(x, \lambda).$

$\therefore$ Strong duality can be interpreted as saddle point property: $\inf_x \sup_{\lambda \geq 0} L(x, \lambda) = \sup_{\lambda \geq 0} \inf_x L(x, \lambda).$

"Saddle point" is. $\tilde{w} \in W, \tilde{z} \in Z$ s.t. $f(\tilde{w}, \tilde{z}) \leq f(\tilde{w}, \tilde{z}) \leq f(\tilde{w}, \tilde{z}) \quad \forall w, z.$

• Complementary Slackness.

assume (x^*, λ^*, μ^*) the optimal points and strong duality holds.

$$\Rightarrow \lambda_i^* f_i(x^*) = 0 \quad \text{for all } i=1, \dots, m!$$

• KKT condition

Assume f_i, h_i differentiable. and strong duality holds.

$\Rightarrow$ Since x^* minimizes $L(x, \lambda^*, \mu^*),$

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^p \mu_i^* \nabla h_i(x^*) = 0.$$

so under 5 conditions are called KKT condition.

1) $f_i(x^*) \leq 0 \quad i=1, \dots, m$

2) $h_i(x^*) = 0 \quad i=1, \dots, p$

3) $\lambda_i^* \geq 0 \quad i=1, \dots, m.$

4) $\lambda_i^* f_i(x^*) = 0 \quad i=1, \dots, m$

5) $\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^p \mu_i^* \nabla h_i(x^*) = 0.$

and we know if (x^*, λ^*, μ^*) are optimal solutions and strong duality holds $\Rightarrow$ satisfy KKT.

Converse -like argument is also true.

if primal problem is convex. and (x^*, λ^*, μ^*) satisfy KKT condition $\Rightarrow$ they are optimal and strong duality holds.

$\therefore$ If Slater's constraint qualification holds. (so strong duality holds) in convex optimization problem

then we could merely solve KKT condition to get optimal solution!

$\Rightarrow$ KKT condition is good because by solving only "equations", we could get optimal solutions.